Extended Abstract: Reproducing and Extending "The Geometry of Truth" to Strategic Deception and Cross-Lingual Transfer

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Abstract

We reproduce "The Geometry of Truth" (Marks & Tegmark, 2023), confirming that Mass-Mean (MM) probes are more causally implicated in representing factual truth than Logistic Regression (LR) probes in LLaMA-2-13B. However, we reveal severe generalization vulnerabilities: probes overfit to structural heuristics unless datasets are perfectly balanced. Extending this work, we evaluate LLaMA-3.1-8B-Instruct, finding improved native decoupling of numerical and factual truth. We also demonstrate cross-lingual transfer, evaluating an English-trained probe on French (97.3%), Spanish (92.0%), and Serbian (76.7%) text. Finally, we test the probe against the Apollo Deception dataset (Goldowsky-Dill et al., 2025). It completely fails to distinguish honest from deceptive prompts, suggesting linear truth probes detect objective semantic correlations rather than strategic intent.

1 Introduction

Understanding LLM truth representation is critical for AI safety. Marks & Tegmark (2023) demonstrate that LLMs possess emergent linear representations of truth, claiming that Mass-Mean (MM) probes identify directions more causally implicated in outputs than Logistic Regression (LR) probes. We reproduce these claims for the EEML Reproduction track. Beyond replication, we critically analyze probe robustness, highlighting generalization vulnerabilities, and extend the methodology to LLaMA-3.1, multi-lingual transfer, and strategic deception detection (Goldowsky-Dill et al., 2025).

2 Reproduction & Implementation Details

We independently reproduced the methodology, extracting last-token hidden state activations from LLaMA-2-13B across factual datasets.

Implementation Challenge: CPU-bound Intel MKL Eigen-decomposition failed on our server's AMD EPYC processors. We resolved this by migrating covariance matrix inversion and PCA entirely to CUDA, bypassing the crash and accelerating extraction.

Reproduction Results: We successfully replicated the central causal claim via activation patching, measuring the Natural Indirect Effect (NIE). As shown in Figure 1(a), adding the MM truth direction to false-statement activations had a substantially higher causal impact on output probabilities ($P(TRUE) - P(FALSE)$) than the LR direction, confirming LR overfits to correlational features while MM captures a more causally relevant geometric center.

3 Criticisms & Generalization Failure

While compelling, our reproduction revealed a severe generalization vulnerability. A probe trained exclusively on the numerical `larger_than` dataset failed entirely to generalize to factual `cities` (accuracy fell to random chance). The original authors mitigated this by perfectly balancing training datasets (combining `larger_than` with `smaller_than`). This highlights a critical limitation:

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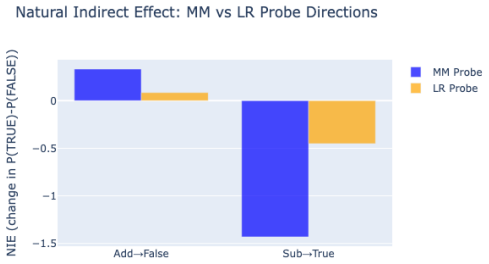
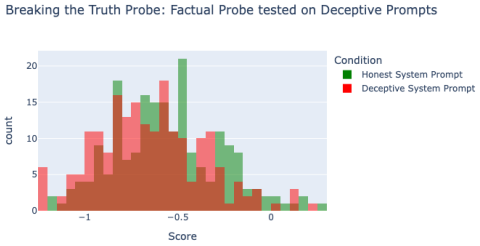
without heavily curated datasets, the method is brittle. At the 13B scale, probes overfit to structural heuristics rather than learning a universal representation of truth.



(a) LLaMA-2 Causal Intervention (NIE). The MM probe shows a significantly higher causal effect than the LR probe.



(b) Breaking the Probe. A factual truth probe completely fails to separate Honest (green) from Deceptive (red) prompts.

Figure 1: Core Reproduction (LLaMA-2) and Deception Extension Results.

4 Extensions: LLaMA-3.1 & Cross-Lingual Transfer

To explore if these vulnerabilities persist, we tested LLaMA-3.1-8B-Instruct. An independent layer sweep identified Layer 10 as the optimal bottleneck. Interestingly, LLaMA-3.1 generalized from math to geography with $> 90\%$ accuracy *without* the dataset balancing tricks required for LLaMA-2 (Appendix A).

Cross-Lingual Transfer: We evaluated the English-trained probe on datasets translated into French, Spanish, and Serbian, achieving zero-shot accuracies of 97.3%, 92.0%, and 76.7%, respectively. This reveals a compelling gradient: truth representations transfer almost perfectly to high-resource languages, but degrade on lower-resource languages like Serbian (though remaining well above chance). Thus, the model's internal concept of truth is largely language-agnostic.

5 Breaking the Probe (Strategic Deception)

Given their sensitivity to semantic heuristics, we hypothesized truth probes would fail to detect actual strategic intent. We evaluated our geography-trained probe against the Apollo Deception dataset (Goldowsky-Dill et al., 2025). As shown in Figure 1(b), the probe completely failed to distinguish between Honest and Deceptive prompts. This critical finding suggests that factual linear probes blindly detect objective semantic correlations regarding prompt content, rather than the model's internal *strategic intent* to deceive.

6 Conclusion

We successfully reproduced the core causal findings of *The Geometry of Truth*. However, our analysis reveals that factual truth probes are brittle and prone to heuristic overfitting. Our extensions to LLaMA-3.1, multi-lingual transfer, and the Apollo Deception dataset highlight a fundamental gap: detecting factual accuracy is not equivalent to detecting strategic deception. These findings warrant further research into intent-based probing.

References

Marks, S., & Tegmark, M. (2023). The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets. *arXiv preprint arXiv:2310.06824*.

Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., & Hobbhahn, M. (2025). Detecting Strategic Deception Using Linear Probes. *Apollo Research*.

A Supplementary Figures and Additional Explorations

This appendix contains additional figures generated during the reproduction (LLaMA-2) and extension (LLaMA-3.1 and Multi-Lingual transfer) phases of the project.

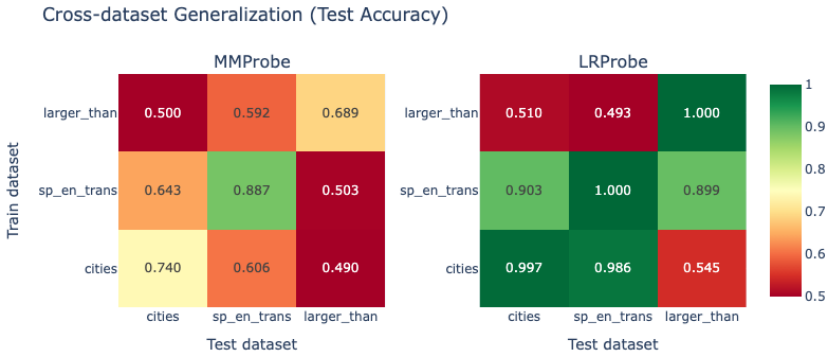


Figure 2: Cross-Dataset Generalization Matrix (LLaMA-2). Notice the failure of probes trained on `larger_than` to generalize to `cities` without dataset balancing.

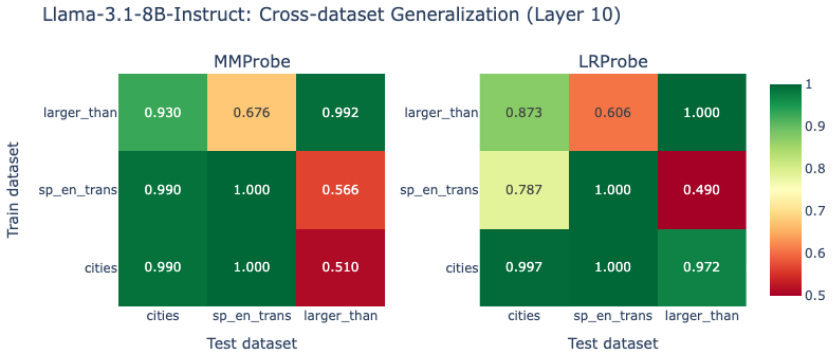


Figure 3: Cross-Dataset Generalization Matrix (LLaMA-3.1). Unlike LLaMA-2, LLaMA-3.1 showed strong off-diagonal generalization (e.g., training on math and testing on geography) without the need for artificial dataset balancing.

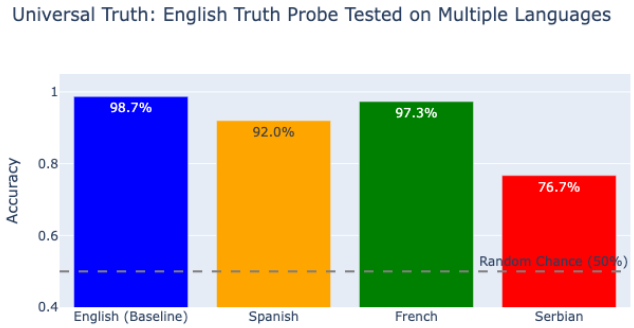


Figure 4: Cross-Lingual Transfer. The English-trained factual probe tested on multiple languages shows near-perfect transfer to high-resource languages (French, Spanish) and partial transfer to lower-resource languages (Serbian).